

# Jorge Domingues Nunes

Senior Fullstack AI Applied Engineer · React & Node.js · Agentic Systems & MCP

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## SUMMARY

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Senior full-stack engineer with 7+ years shipping production web applications and 3+ years building AI-driven products end-to-end. I build and maintain scalable apps in React / Next.js and Node.js / TypeScript (plus Python/FastAPI), own the full SDLC — design, coding, testing, debugging, refactoring and documentation — and back it with disciplined testing: unit coverage, functional and integration suites, and TDD on core paths. I work natively with AI coding assistants (Claude Code, Cursor, GitHub Copilot, Codex, ChatGPT, Google Antigravity, Devin, VS Code, Windsurf) as a daily workflow, not just a tool, and validate AI-generated code for production readiness, security and maintainability. I design the guardrails — testing frameworks, CI/CD gates and human-in-the-loop review — that make AI-assisted delivery fast and safe, and I mentor engineers on adopting it responsibly.

## EXPERIENCE

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### Founder & AI Solutions Architect — Exponential

Nov 2025 – Present

*Agentic AI marketplace & orchestration platform, built end-to-end (React, Node/FastAPI, Flutter, infra).*

- **Full-stack product:** React dashboard, FastAPI backend (~200 endpoints), Flutter iOS/Android app, and an embeddable chatbot widget with session persistence, rate limiting and domain allowlisting.
- **Testing & quality:** Unit suites with coverage thresholds, integration tests against live services, and TDD on the orchestration core, all enforced through CI before deploy.
- **AI-assisted delivery:** Day-to-day pair-programming with Claude Code, Cursor, GitHub Copilot, Codex, ChatGPT, Google Antigravity, Devin, VS Code and Windsurf across backend, infra and mobile; AI output reviewed and tested before merge.
- **Agentic orchestration:** Multi-agent runtime on LangChain / LangGraph with planner/executor patterns and MCP-style tool integrations (Terraform, Docker, UI generation), with Ollama-powered local inference.
- **RAG, memory & retrieval:** WebRAG over ChromaDB with hybrid search plus Tavily web search, per-user conversation memory, and prompt/response caching — ~90% LLM cost reduction on repeat workloads.
- **Infra:** Kubernetes with custom CRDs for agent pods, Helm, Kustomize, Terraform across Docker / AWS / GCP / Azure, Ansible for post-deploy config.

### Principal AI Engineer — Keyprog ([keyprog.pt](https://keyprog.pt))

Jan 2024 – Oct 2025

*LLM-powered e-commerce SaaS, Directus tooling & MCP · Remote*

- **Full-stack delivery:** Shipped a production e-commerce platform end-to-end: Node.js / TypeScript and FastAPI services, React front-ends, and a Directus headless CMS as the content and access layer.
- **Directus headless Visual Editor:** Integrated the Directus Visual Editor ([@directus/visual-editing v2](https://github.com/directus/directus-visual-editing)) into a Next.js 15 / React 19 storefront for in-context, live editing of headless content — letting editors update production pages directly against the CMS.
- **MCP server (published on npm):** Designed and shipped [@staminna/directus-mcp-server](https://github.com/staminna/directus-mcp-server) (v12) — 29+ tools, TypeScript, WebSocket and full Directus API coverage — so AI coding agents and product agents can query, automate, create API endpoints, create SQL databases and relationships, build plugins and reason over any Directus backend in natural language.
- **RAG & prompt engineering:** Retrieval pipelines over product and content data with Tavily web search and structured prompting.

- **Testing & delivery:** Unit and integration coverage with a TDD approach on critical flows, Docker and GitHub Actions CI/CD; mentored engineers on LLM integration patterns and prompt/eval conventions.



### Senior AI/ML Engineer — Co2Offset.ai

Apr 2023 – Dec 2024

*Forest carbon-capture SaaS · Pombal, Portugal*

- SaaS measuring carbon captured in forests from satellite and ground data, rewarding landowners for protection.
- **Full-stack & ML pipeline:** Full role-based permissions system for user profiles and policies; AWS, MB WAY and Moloni integrations, with Python computer-vision microservices behind FastAPI; partnered with ESA on geospatial data services.
- **Outcome:** Led the pipeline audit with Bureau Veritas, securing the scientific-reliability certification that unlocked €2.5M of business.
- **Data & platform:** PostgreSQL with PostGIS for geospatial queries, MongoDB for unstructured layers, Redis caching; Kubernetes-deployed services with auto-scaling, health probes and multi-env Terraform.



### Technical Consultant — European Central Bank

Mar 2020 – Mar 2023

*Main website design team · Agile environment · Frankfurt, Germany*

- Core member of the ECB main-website design team, delivering the public site for one of the world's largest central banks through agile, sprint-based collaboration.
- Built a JavaScript migration tool converting legacy HTML and JS to the new design's IDs and classes.
- Drove a multilingual website redesign serving millions of articles across 24 languages.
- Built an Elasticsearch-based search engine indexing internal CMS and public-site content in 24 languages, serving millions of pages.
- Intranet moderator; experienced working in restricted, security-controlled environments.
- Owned release engineering, end-to-end testing, and security hardening (CSP, SSL, Apache server configuration for 12 team members with Virtual Hosts).



### Senior Drupal Developer — European Commission

Apr 2018 – Jan 2020

*Public-facing platforms · Agile environment · Brussels / Luxembourg*

- Member of the main-site redesign team delivering public-facing platforms for the European Commission in an agile, sprint-driven workflow.
- Designed the relational data model serving tens of thousands of users.
- Built CI/CD pipelines with Jenkins, GitHub Actions and Drone, with automated tests in Jest.
- Led several Drupal migrations across the platform.

## EDUCATION

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### Computer Graphics & Multimedia Engineering

09/2003 – 06/2006

*Instituto Politécnico de Viana do Castelo — Escola Superior de Tecnologia e Gestão*

- Programming in C, C++ and Java with a focus on object-oriented design and real-time systems.
- Real-time 3D graphics programming with OpenGL and DirectX (hardware-accelerated GPU graphics), plus AutoCAD, UML and UX/UI design principles.
- Mathematics, Linear Algebra and Calculus; Information Systems Programming; Programming & Algorithms (programming-logic foundations).
- IT Security, Communication Networks and Technical English.

### Computer Technician — Informatics Technological Course

09/2001 – 06/2003

*Escola Secundária de Pombal, Portugal*

- Programming foundations: Pascal, C, Visual Basic, JavaScript and ActionScript; programming logic, algorithms and software development.
- Databases: MSSQL and SQL fundamentals — table design, queries and relationships; introduction to backend concepts.
- Systems & networking: installing and configuring Windows / Linux, LAN and Internet protocols, router/switch configuration, and web/file/database servers.
- IT security: network security concepts, safe use of systems and networks, malware awareness, and backup & data protection.

## SELECTED PROJECTS

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- [CI/CD Pipeline MCP server](#) A Model Context Protocol (MCP) server that provides AI assistants with **unified awareness** of local and remote Docker environments. Works with Claude Desktop, Cursor, Windsurf, and any MCP-compatible IDE.
- [RAG-PDF](#) (LangGraph + LangChain) — retrieval-augmented PDF Q&A built as a LangGraph state machine over LangChain retrievers.
- [@stamina/directus-extension-typeform](#) — Built and published a bidirectional Typeform integration with one-shot form import, Flow operations over the full Typeform API, and webhook sync with auto-mapped collections.
- [Sentiment Analyzer](#) (Go) — CLI tool for sentiment analysis via the Hugging Face API.

## TECH STACK

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- **Frontend / mobile:** React, Next.js, TypeScript, MUI, Flutter / Dart
- **Backend & data:** Node.js, FastAPI, Python, REST & WebSocket APIs, PostgreSQL / PostGIS, Redis, MongoDB
- **Testing & QA:** Jest, Pytest, unit / functional / integration testing, coverage gates, TDD, MLflow evals
- **AI coding assistants:** Claude Code, Cursor, GitHub Copilot, Codex, ChatGPT, Google Antigravity, Devin, VS Code, Windsurf — daily across backend, infra and mobile
- **Agents & LLM tooling:** LangChain, LangGraph, MCP, Tavily, OpenAI, Anthropic, Gemini, Ollama (Mistral / Llama 3 / Qwen / DeepSeek)

- **Retrieval:** ChromaDB, Qdrant, Pinecone, hybrid search, embeddings
- **Infra & ops:** Docker, Kubernetes, Helm, Terraform, GitHub Actions, AWS / GCP / Azure

## CORE COMPETENCIES

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- **Full-stack web engineering:** React / Next.js, Node.js / TypeScript, FastAPI; REST & WebSocket APIs, server components, MUI, responsive and embeddable UIs — frontend, backend, and the data layer under them.
- **AI-assisted development across the SDLC:** Claude Code, Cursor, GitHub Copilot, Codex, ChatGPT, Google Antigravity, Devin, VS Code and Windsurf used daily to design, generate, test, debug, refactor and document; I write prompts, workflows and agent instructions that measurably raise team throughput.
- **Testing, quality & guardrails:** Unit testing with coverage gates, functional and integration suites, and TDD; CI/CD pipelines and human-in-the-loop validation that vet AI-generated code for correctness, security and maintainability before it ships.
- **Agentic workflows & MCP:** Designed and published MCP servers on npm and the Directus Marketplace; multi-agent orchestration with LangGraph / LangChain (planner/executor, tool use, memory) shipped to real users.
- **RAG & LLM tooling:** Hybrid search and vector stores (ChromaDB, Qdrant, Pinecone), Tavily-powered web search and retrieval, prompt engineering, Knowledge Base wired into production surfaces.

**Data & cloud:** PostgreSQL / PostGIS, Redis, MongoDB; Docker, Kubernetes, Terraform and GitHub Actions CI/CD across AWS / GCP / Azure.

## CONTINUOUS LEARNING

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LLM fine-tuning, RAG architectures, prompt engineering and vector databases; agentic workflow patterns (planner/executor, tool use, multi-agent coordination, skills); testing strategy and validation for AI-assisted development; Terraform & Kubernetes operations; Flutter cross-platform architecture.